Chapter 1

Week 13-19 October

1.1 Introduction

In most of the course we'll handle wireless communications, as they're much more difficult than wired communications. In this course we'll stay at the physical layer. We'll see:

- **Channel models**: taking radio communications and making a mathematical model for them. We'll find an accurate model and then we'll apply a tradeoff to it, simplifying it so that it is manageable.
- **OFDM**: Orthogonal Frequency Division Multiplexing.
- **MIMO**: we have multiple antennas at the transmitter and multiple antennas at the receiver; we'd like to see how much this improves communications. The book he selected this year handles fundamentals of MIMO communications. MIMO is the latest and greatest achievement, he chose a book which goes from the basics. We might have single user MIMO or even multiple user MIMO: we're communicating to multiple phones through multiple antennas.

We can ask ourselves: what is the best that we can do? This is what our approach will be, so that we can know the boundaries. We'll follow the slides from the author of the book, hopefully they'll be good. We'll also have exercises with solutions: we can look at them from cambridge.org , specifically from the page of the book.

1.2 Primer

The first chapter is a primer in information theory and MSE estimation; we'll go fast as it should summarize stuff we know, as mutual information between two variables. !!: the signals are always complex-valued and the constellations are always normalized to have unit power. ∞-QAM: we increase the number of points making them denser inside a specific set.

1.2.1 Complex symmetric random variables, ∞-modulations

We'll start from chapter 1 of the book (000): *primer of information theory and MSE.*

We should already know this stuff, but we'll cover it in order to have a common starting ground.

∗Prof's backstory moment∗: In the end, he studies **physical layer** and that's it.

We ask ourselves: "**what is information**? When can we say that someone is giving us information?". ∗Professor begins ranting about information∗: information is related to **probability**! The more unlikely something is, the more informative it is!

Today we'll talk about **mutual information** between two random variables: the information a variable can supply to another variable.

We'll talk about **channel capacity**: the capacity at which a channel can convey information; it answers to "how much information can I send through this channel, reliably?". By reliably we mean that we can lower the probability of error as much as we want.

Binary coding and **decoding** are near optimum, but there can be other ways, which could be better: we'll show that binary is indeed **optimum**.

LMMSE is optimum when the signal and noise are **Gaussian**.

Let's start from **signals**; in the book, signals are normally **complex**, unless we specifically say otherwise. Normally **complex random variables** are **complexly symmetric**:

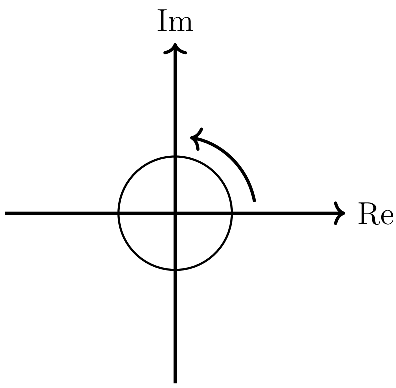
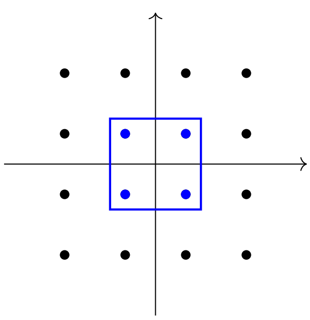


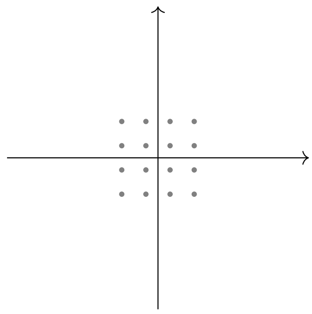
Figure 1.1: Rappresentazione del piano Im-Re con una PFD simmetrica centrale e freccia di rotazione.

About **M-PSK** and **M-QAM**: M is usually a **power of 2**.

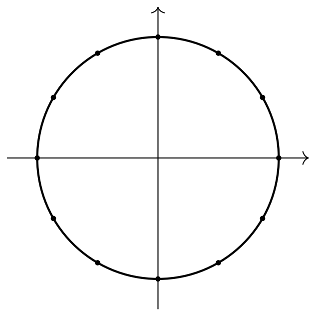
Let's see **16-QAM**:



The power of the ones in blue is lower than the power of the other signals. In the book they always keep the **same variance of the constellation**: the book squeezes the values so that the **power stays constant**.



The book also has **inf-PSK** and **inf-QAM**: We take the **normal constellation** and we keep **adding points**.



$\infty - \textbf{PSK}$

It's a useful way to see what happens in the **asymptotic regime**. Also, the **PDF of a complex Gaussian variable exists** (see notes).

If σ^2 is the variance, the *Standard deviation* is $\sqrt{\sigma^2} = \sigma$

NOT $\pm\sigma^2$!

We'll try and be as precise as possible: there is no **negative standard deviation**! As an example:

$$f(x) = \sqrt{x} \in \mathbb{R} \text{ is def. only for } x \geq 0$$
$$x^2 = 16 \text{ has two sol.}$$

What about information? There have been many results, dating back to **Shannon**! (See pag 7 of the slides btw)

1.3 Discrete entropy, proper signals

1.3.1 Entropy

Entropy of a discrete-time random variable

$$\begin{aligned}\mathcal{H}(x) &= -\sum_x p_x(x)\log_2 p_x(x)\\ &= -\mathbb{E}[\log_2 p_x(x)].\end{aligned}$$

Joint entropy

$$\begin{aligned}\mathcal{H}(x_0,x_1) &= -\sum_{x_0}\sum_{x_1} p_{x_0x_1}(x_0,x_1)\log_2 p_{x_0x_1}(x_0,x_1)\\ &= -\mathbb{E}[\log_2 p_{x_0x_1}(x_0,x_1)].\end{aligned}$$

> **Nonnegative function that quantifies the amount of uncertainty in discrete valued RV x**

Conditional entropy

$$\mathcal{H}(x|y) = -\sum_x\sum_y p_{xy}(x,y)\log_2 p_{x|y}(x|y).$$

Entropy connections

$$\mathcal{H}(x,y) = \mathcal{H}(x) + \mathcal{H}(y|x).$$

The entropy is used in statistical physics.

Joint entropy:

$$H(x_0,x_1) = \sum_{x_0}\sum_{x_1} p_{x_0,x_1}(x_0,x_1)\log_{1/2} p_{x_0,x_1}(x_0,x_1)$$

△ **Diff. inform. <u>can</u> be negative**

Conditional entropy:

$$H(x,y) = \sum_x\sum_y p_{x,y}(x,y)\log_2(p_{x|y}(x|y))$$

From this, we can use the Bayes theorem to derive:

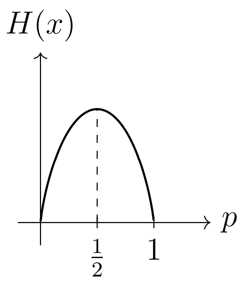
$$\begin{aligned}H(x,y) &= H(x|y) + H(y)\\ &= H(y|x) + H(x)\end{aligned}$$

Example: We have a Bernoulli variable

$$X = \{0, 1\} \qquad P = \{p, 1 - p\}$$

$H(X)$ will be ≤ 1, where the equality holds if $p = 1/2$.

If we plot it, we get:

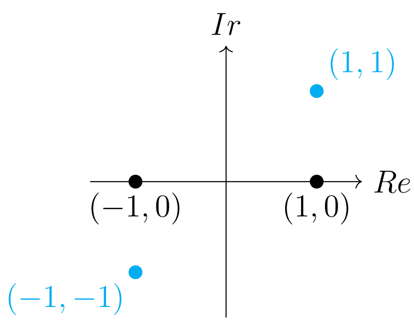


We get maximum information when we flip a ***fair*** coin!
If the coin always flips heads or tails, we get no information!

We could say that the messier the information, the most we get out of it.

Prof forgot stuff about signals: we're always talking about "proper" signals; the real and imaginary part of those numers are uncorrelated!

Properness is preserved by affine transformations. An intuitive reason why we consider only proper complex numbers is that if we have real and imaginary parts of a random variable:



$(-1, -1)$ **This is a fake complex variable; we want to avoid these cases.**

To elaborate further:

Proper complex

A random variable (or a random process/random vector) is said to be proper complex if it satisfies **Properness**:

Properness

A complex random variable is said to be **proper** if

$$E[x^2] = E[x]^2$$

holds. Properness requires $\mathrm{Re}[x], \mathrm{Im}[x]$ to be uncorrelated and have the same variance.

A complex random vector is said to be **proper** if

$$E[\mathbf{x}\mathbf{x}^T] = E[\mathbf{x}]E[\mathbf{x}]^T$$

holds, which is equal to asking that $\mathrm{Re}[\boldsymbol{x}], \mathrm{Im}[\boldsymbol{x}]$ have identical Covariance matrices and cross-covariance equal to 0.

Properness is preserved in affine transformations!

1.3.2 Entropy rate

Let's say we have a sequence of discrete random variables indexed by time, $x_0, \ldots, x_{N-1}$. If they are **IID**, then the entropy of the whole process grows linearly with N at a rate $\mathcal{H}(x_0)$.

In general, a process grows according to the so-called *entropy rate*

$$\mathcal{H} = \lim_{N\to+\infty} \frac{1}{N}\mathcal{H}(x_0, \ldots, x_{N-1}). \tag{1.32}$$

If the process is **stationary**, we can show that the entropy rate is equal to

$$\mathcal{H} = \lim_{N\to+\infty} \mathcal{H}(x_N|x_0, \ldots, x_{N-1}). \tag{1.33}$$

All of this can be extended for continuous processes to differential entropy.

A process is said to be **nonregular** if its present value is perfectly predictable from noiseless observations of the entire past, while the process is **regular** (which happens if $\mathfrak{h} > -\infty$) if its present value cannot be perfectly predicted from noiseless observations of the entire past.

1.3.3 Differential entropy

Differential entropy:

$$h(x) = \int f_x(x) \log_{1/2} f_x(x)\, dx = \mathbb{E}\left[\log_{1/2} f_x(x)\right]$$

For vectors:

$$h(\underline{x}) = \mathbb{E}\left[\log_{1/2} f_{\underline{x}}(\underline{x})\right]$$

The differential entropy rate does not measure the information contained in a random variable. If we only have a random variable, we only have a shot. In order to consider a **rate**, we consider a **random process**: in this case, we have a random variable every certain amount of time: we can say that we have a certain amount of information per second!

A binary random process with $p = 0.5$ which gives a value every second, gives 1 bit/s.

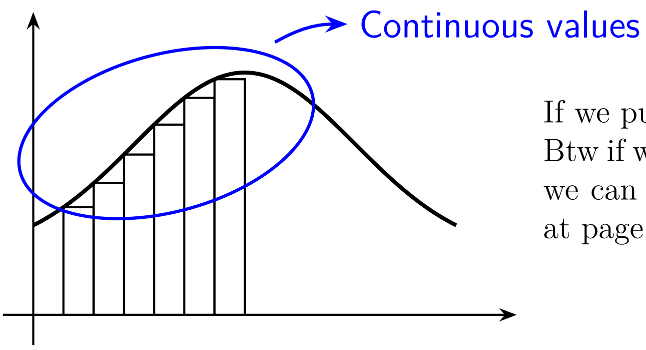
So the rate does not measure the information contained in a random variable. We also need to keep in mind that the differential entropy can be negative.

It's easy to prove that a **Gaussian distribution maximizes the differential entropy**, for a fixed variance.

The differential entropy of a b bit quantization of a random variable x is $h(x) + b$ (what happens if we quantize more and more finely?)

If we discretize a variable in b bits, then what we obtain is another random variable. What if we want to compute the differential entropy of that random variable?

Why do we use differential entropy even though we discretized? We just elongate it:



If we push b to inf, it diverges!
Btw if we want a reference for information theory, we can use the book "Cover & Thomas" (this is at page 247).

Pag 10 of the slides: how to calculate the differential entropy of a Gaussian complex vector.

Example 1.6 (Differential entropy of a complex Gaussian vector)

Let $\mathbf{x} \sim \mathcal{N}_{\mathcal{C}}(\boldsymbol{\mu}, \boldsymbol{R})$. From (C.15) and (1.22).

$$\begin{aligned}
h(\mathbf{x}) &= -\mathbb{E}[\log_2 f_{\mathbf{x}}(\mathbf{x})] && (1.23)\\
&= \log_2 \det(\pi \boldsymbol{R}) + \mathbb{E}[(\mathbf{x}-\boldsymbol{\mu})^* \boldsymbol{R}^{-1}(\mathbf{x}-\boldsymbol{\mu})] \log_2 e && (1.24)\\
&= \log_2 \det(\pi \boldsymbol{R}) + \mathrm{tr}(\mathbb{E}[(\mathbf{x}-\boldsymbol{\mu})^* \boldsymbol{R}^{-1}(\mathbf{x}-\boldsymbol{\mu})]) \log_2 e && (1.25)\\
&= \log_2 \det(\pi \boldsymbol{R}) + \mathrm{tr}(\mathbb{E}[\boldsymbol{R}^{-1}(\mathbf{x}-\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu})^*]) \log_2 e && (1.26)\\
&= \log_2 \det(\pi \boldsymbol{R}) + \mathrm{tr}(\boldsymbol{R}^{-1} \underbrace{\mathbb{E}[(\mathbf{x}-\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu})^*]}_{\boldsymbol{R}}) \log_2 e && (1.27)\\
&= \log_2 \det(\pi \boldsymbol{R}) + \mathrm{tr}(\boldsymbol{I}) \log_2 e && (1.28)\\
&= \log_2 \det(\pi e \boldsymbol{R}), && (1.29)
\end{aligned}$$

where in (1.25) we used the fact that a scalar equals its trace, while in (1.26) we invoked the commutative property in (B.26).

Note: $\det(\exp(\mathbf{A})) = \exp(\mathrm{tr}(\mathbf{A}))$

Example 1.5 (Differential entropy of a complex Gaussian scalar)

Let $x \sim \mathcal{N}_{\mathcal{C}}(\mu, \sigma^2)$. Invoking the PDF in (C.14),

$$h(x) = \mathbb{E}\left[\frac{|x-\mu|^2}{\sigma^2}\log_2 e + \log_2(\pi\sigma^2)\right] = \log_2(\pi e \sigma^2).$$

(Scalar result)

Equivalent scalar result

Professor freaks out over the fact that we don't know what the trace function is. The trace of a matrix is the sum of the diagonals. If we want to compute the sum of the eigenvalues, we can just compute the trace of a matrix.

btw $\quad \det(\exp(\underline{A})) = \exp(\mathrm{tr}(\underline{A}))$

So we have that

$$h(\underline{x}) = \log_2(\det(\pi e \underline{R}))$$
$$h(x) = \log_2(\pi e \sigma^2)$$

At least not for a Gaussian R.V.!

The differential entropy DOES NOT depend on the mean of the variable!

Another very important thing is that the differential entropy is a property of the distribution of a random variable, not of the random variable itself. Two random variables having the same distribution will have the same random variable.

$$R.V. \text{ is a map } \begin{tikzcd} \Omega \arrow[r] \arrow[d] & \mathcal{S} \arrow[d] \\ \textit{events} & \textit{probability} \end{tikzcd}$$

Elaborating a bit on the computation of the differential entropy: we have that

$$\begin{aligned} h(\boldsymbol{x}) &= -\mathbb{E}[\log_2 f_{\boldsymbol{X}}(\boldsymbol{x})] \qquad (1.23) \\ &= -\mathbb{E}\left[\log_2\left(\frac{1}{\det(\pi \boldsymbol{R}_x)}\exp(-(\boldsymbol{X}-\boldsymbol{\mu}_x)^*\boldsymbol{R}_x^{-1}(\boldsymbol{X}-\boldsymbol{\mu}_x))\right)\right] \\ &= \log_2(\det(\pi\boldsymbol{R}_x)) - \mathbb{E}\left[\frac{\ln(\exp(-(\boldsymbol{X}-\boldsymbol{\mu}_x)^*\boldsymbol{R}_x^{-1}(\boldsymbol{X}-\boldsymbol{\mu}_x)))}{\ln 2}\right] \\ &= \log_2(\det(\pi\boldsymbol{R}_x)) - \mathbb{E}\left[\frac{-(\boldsymbol{X}-\boldsymbol{\mu}_x)^*\boldsymbol{R}_x^{-1}(\boldsymbol{X}-\boldsymbol{\mu}_x)}{\ln 2}\right] \\ &= \log_2(\det(\pi\boldsymbol{R}_x)) + \mathbb{E}\left[(\boldsymbol{X}-\boldsymbol{\mu}_x)^*\boldsymbol{R}_x^{-1}(\boldsymbol{X}-\boldsymbol{\mu}_x)\right]\log_2 e \qquad (1.24) \end{aligned}$$

and now we observe that $(\boldsymbol{X}-\boldsymbol{\mu}_x)^*\boldsymbol{R}_x^{-1}(\boldsymbol{X}-\boldsymbol{\mu}_x)$ actually is a scalar (given that $\boldsymbol{x}$ is a $N\times 1$ vector, we're multiplying $(1\times N)\cdot(N\times N)\cdot(N\times 1)\to 1$): we can then exploit the fact that a scalar is equal to its trace together with the fact that the trace is a scalar operation to get

$$
\begin{aligned}
&= \log_2(\det(\pi\boldsymbol{R}_x)) + \mathrm{Tr}\left[\mathbb{E}\left[(\boldsymbol{X}-\boldsymbol{\mu}_x)^*\boldsymbol{R}_x^{-1}(\boldsymbol{X}-\boldsymbol{\mu}_x)\right]\right]\log_2 e && (1.25)\\
&= \log_2(\det(\pi\boldsymbol{R}_x)) + \mathbb{E}\left[\mathrm{Tr}\left[(\boldsymbol{X}-\boldsymbol{\mu}_x)^*\boldsymbol{R}_x^{-1}(\boldsymbol{X}-\boldsymbol{\mu}_x)\right]\right]\log_2 e \qquad \text{Linearity}\\
&= \log_2(\det(\pi\boldsymbol{R}_x)) + \mathbb{E}\left[\mathrm{Tr}\left[\boldsymbol{R}_x^{-1}(\boldsymbol{X}-\boldsymbol{\mu}_x)(\boldsymbol{X}-\boldsymbol{\mu}_x)^*\right]\right]\log_2 e \qquad \text{Ciclicity}\\
&= \log_2(\det(\pi\boldsymbol{R}_x)) + \mathrm{Tr}\left[\boldsymbol{R}_x^{-1}\boldsymbol{R}_x\right]\log_2 e\\
&= \log_2(\det(\pi\boldsymbol{R}_x)) + \mathrm{Tr}\left[\boldsymbol{I}\right]\log_2 e && (1.28)
\end{aligned}
$$

if then we consider $\boldsymbol{R}_x, \boldsymbol{I}$ as $N\times N$ matrices (which they are, this is just needed to say that N is their measure), we'll have

$$
\begin{aligned}
&= \log_2(\det(\pi\boldsymbol{R}_x)) + N\log_2 e\\
&= \log_2(e^N\det(\pi\boldsymbol{R}_x))
\end{aligned}
$$

and, finally, for a matrix $\boldsymbol{A}$ of dimensionality $N\times N$ it holds that

$$a^N\det(\boldsymbol{A}) = \det(a\boldsymbol{A}),$$

so in our situation this is

$$= \log_2(\det(\pi e\boldsymbol{R}_x)). \tag{1.29}$$

Whew, that was a lot!

Finally, we can simplify this for the scalar case as the determinant of a scalar is the scalar itself, getting that

$$h(x) = \log_2(\pi e\sigma^2).$$

> Btw the B.26 relation that they mentioned is
>
> $$\mathrm{tr}(\boldsymbol{AB}) = \mathrm{tr}(\boldsymbol{BA}), \tag{B.26}$$
>
> which was here applied to $\boldsymbol{A} = (\boldsymbol{x}-\boldsymbol{\mu})^*\boldsymbol{R}^{-1}, \boldsymbol{B} = (\boldsymbol{x}-\boldsymbol{\mu})$.

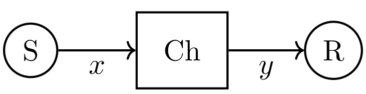
1.4 Information divergence, mutual information

We have that the Kullback-Leibler divergence is defined as

$$\mathcal{D}(p||q) = \begin{cases} \sum\limits_x p(x)\log_2\left(\frac{p(x)}{q(x)}\right) & \text{discrete } x\\ \int p(x)\log_2\left(\frac{p(x)}{q(x)}\right) & \text{continuous } x \end{cases} \tag{1.34+1.36}$$

this is also called the *relative entropy*.

Another way to measure such thing is the Earth-Mover distance.



The mutual information quantifies the reduction in uncertainty between two random variables if one of them is observed.

Discrete:

$$\mathrm{I}(s;y) = H(s) - H(s|y) = H(y) - H(y|s)$$

Between s and y

$$= D(p_{sy}||p_s p_y)$$

$$= \sum_s \sum_y p_{sy}(s,y) \log_2 \left(\frac{p_{sy}(s,y)}{p_s(s) p_y(y)} \right)$$

Cont:

$$\begin{aligned} &= \mathbb{E}[\text{information density}] \equiv \mathbb{E}[i(s;y)] \\ \mathrm{I}(s;y) &= h(s) - h(s|y) = h(y) - h(y|s) \\ &= D(f_{sy}||f_s f_y) \\ &= \iint f_{sy}(s,y) \log_2 \left(\frac{f_{sy}(s,y)}{f_s(s) f_y(y)} \right) ds\, dy \end{aligned}$$

He talked about the Landau symbols ; he didn't say anything relevant really.

1.4.1 Power series expansions

Useful means to approximate functions especially low or high SNR.

- **Zero order expansion about 0**

$$\boxed{f(x) = f(0) + \mathcal{O}(x)}$$

- **First order expansion about 0**

$$\boxed{f(x) = f(0) + f'(0)x + \mathcal{O}(x^2)}$$

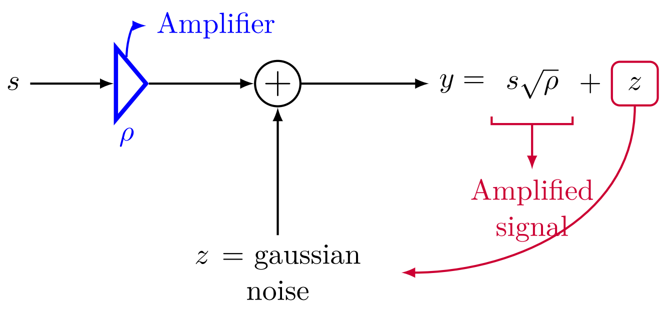
- **First order expansion about Infinity**

$$\boxed{f(1/x) = f(0) + \frac{f'(0)}{x} + \mathcal{O}\left(\frac{1}{x^2}\right)}$$

Mathematica snippet:

```
In[11]:= Series[f[x], {x, 0, 2}]
Out[11]= f[0] + f'[0]x + 1/2 f''[0]x^2 + O[x]^3
```

Important! → SL. $\boxed{16}$



We can ask ourselves: how much is the gaussian noise corrupting what I'm sending?

We amplify ρ (which is $\sqrt{\rho}$ in power) $\Rightarrow$ we get an ampl. of $\sqrt{\rho}$ on power!

1.4.2 Mutual information of a Gaussian scalar through an AWGN channel

We were calculating the **mutual information** for a Gaussian complex scalar.

We talked about the relation $y = \sqrt{\rho}s + z$. We have that s, z are **independent**. It might seem weird that we're putting noise in a channel, but it will make sense later.

Our constellation input always will have $\sigma^2 = 1$, such that $s \sim \mathcal{N}_{\mathcal{C}}(0, 1)$; we'll also have a noise $z \sim \mathcal{N}_{\mathcal{C}}(0, 1)$. We have that y is made of a sum of Gaussian independent random variables, and will thus be a Gaussian random variable, with a variance given by a sum of variances: we'll have

- $y \sim \mathcal{N}_{\mathcal{C}}(0, 1 + \rho)$
- $y|s \sim \mathcal{N}_{\mathcal{C}}(\sqrt{\rho}s, 1)$: this is y with s being fixed; we fix the $\sqrt{\rho}s$ in y's equation. We now want to calculate the mutual information between the input and the output of the signal.

We know the differential entropy of a Gaussian scalar and we know that

$$
\begin{aligned}
\mathcal{I}(\rho) &= I(s; \sqrt{\rho}s + z) \\
&= h(\sqrt{\rho}s + z) - h(\sqrt{\rho}s + z|s) \\
&= h(\sqrt{\rho}s + z) - h(z) \\
&= \log_2(\pi e(1 + \rho)) - \log_2(\pi e) \\
&= \log_2(1 + \rho). \qquad (1.51)
\end{aligned}
$$

Relevant comments:

- In $h(\sqrt{\rho}s + z|s) = h(z)$, we know that we've already **fixed** s; s multiplied by whichever scalar won't bring any extra information than z, as we already know what s is; for this reason, the differential entropies are the same.

If we then consider $\rho \to 0$, we have

$$\log_e(1+x) \xrightarrow{x \to 0} x - \tfrac{1}{2}x^2 + \mathcal{O}(x^3)$$

So we need to **convert to base** e and we'll then be able to use the expansion:

$$\log_2(1+\rho) = \left(\rho - \tfrac{1}{2}\rho^2\right)\log_2 e + \mathcal{O}(\rho^3)$$

And **for large** ρ we'll then have

$$\log_2(1+\rho) \xrightarrow{\rho \to \infty} \log_2(\rho) + \mathcal{O}(1/\rho)$$

■ **Note**

This is **crucially important** as it allows to model an AWGN channel! It allows us to define the **capacity** as

$$\frac{1}{2}\log(1+\mathrm{SNR})$$

(we don't have 1/2 since this is complex; we'd have 1/2 if it wasn't complex!).